Harsh Shrishrimal

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EDUCATION

University of Maryland **College Park, MD**
M.S. in Data Science *Aug 2024 - May 2026*

- **GPA:** 3.78/4.0
- **Related Coursework:** Probability & Statistics, Machine Learning, Cloud Computing

Savitribai Phule Pune University **Pune, India**
B.E in Computer Science *Aug 2019 - Jun 2023*

- **GPA:** 7.72/10.0
- **Related Coursework:** Database Management Systems, Business Intelligence, Data Science and Big Data

EXPERIENCE

Converge Technology Solutions **New York, NY**
Artificial Intelligence Intern *Jun 2025 – Aug 2025*

- Built automated prompt injection testing pipelines for LLM-based chatbots, executing 150+ attacks to identify vulnerabilities.
- Implemented DSPy-based prompt optimization across 4 LLMs (GPT, Mistral, LLaMA, Qwen) to improve vulnerability detection.
- Automated attack execution using Docker, Redis, Celery, and Flower monitoring, reducing manual testing effort by 20%.
- Integrated and tested multiple AI chatbot endpoints through automated security pipelines.
- Improved backend orchestration, debugging, and MLflow-based experiment tracking workflows, reducing validation time.

Shanti Technologies **Pune, India**
Machine Learning Engineer *Jul 2023 – Jun 2024*

- Trained TensorFlow demand forecasting models via seasonality and holiday features, reducing inventory stockouts by 18%.
- Developed dynamic pricing algorithms (Python, Flask APIs), increasing campaign revenue by 10%.
- Built scalable ETL pipelines with Apache Airflow and GCP, processing 2M+ daily records with <25-minute latency.
- Designed collaborative-filtering recommendation systems (Scikit-learn, TensorFlow), boosting loyalty engagement by 15%.
- Streamlined ML deployment workflows using Docker, Git, and Jupyter, improving model reliability and release efficiency.

PROJECTS

AI Hedge Fund – Multi-Agent Stock Analysis Platform

- Built a multi-agent stock analysis system (LangGraph, Gemini) with 5 AI agents delivering structured trade signals with confidence scores and position sizing.
- Implemented a ChromaDB RAG pipeline on SEC 10-K/10-Q filings with hallucination grading, achieving reliable source attribution across 4 fault-tolerant data sources.

MedTeller – Vision-Language Radiology Report Generator

- Architected a Vision Transformer + GPT-2 multimodal pipeline (Python, PyTorch) to generate structured radiology reports from 7.4K chest X-rays using cross-modal visual-text alignment.
- Implemented custom preprocessing, tokenization, and attention fine-tuning to improve clinical text coherence and image-report consistency.

CAFB AI Document Generator

- Built a Retrieval-Augmented Generation (RAG) system using CLIP embeddings and Pinecone vector search, achieving 90%+ semantic retrieval accuracy across large-scale document corpora.
- Developed end-to-end Python pipelines for embedding generation, indexing, and prompt-grounded response generation.

PUBLICATIONS

- Co-authored "Review on Handwritten Character Recognition", published in IJIRCCE, Vol. 11, Issue 3, March 2023, DOI: 10.15680/IJIRCCE.2023.1103081.
- Built a LeNet-5 CNN model on the MNIST dataset (70K images) for handwritten digit recognition, achieving 98.23% accuracy with a real-time prediction GUI.

SKILLS & CERTIFICATIONS

Programming Languages: Python, SQL, Java, R, C++, JavaScript, HTML
Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Tableau
Machine Learning & Deep Learning: Scikit-learn, TensorFlow, PyTorch, Keras, OpenCV, DSPy, DQN, CLIP, MLflow
LLMs & Generative AI: LangChain, LangGraph, DSPy, RAG, ChromaDB, Pinecone, Google Gemini, OpenAI, Groq
Cloud & Big Data: Google Cloud (BigQuery, Cloud Functions), AWS, Apache Airflow, Hadoop, Spark
Web & Backend Development: FastAPI, Django, Flask, Celery, Redis, Flower, REST APIs, WebSockets, Next.js
Databases: PostgreSQL, SQLite, BigQuery
Tools & Platforms: Docker, Git, Jupyter, VS Code, Jira, MS Excel, Powerpoint
Certifications: Artificial Intelligence Fundamentals – IBM, Big Data 101 – IBM, Data Analytics & Visualization – Accenture, Databases for Developers – Oracle, Hadoop 101 – IBM